Ex. No.:1	Building E-commerce Chatbot	URK23CS1167
06.08.2026		

Aim:

To build an AI-powered E-Commerce Chatbot using Retrieval-Augmented Generation (RAG) with Ollama.

Problem Statement:

Design and develop a modular Retrieval-Augmented Generation (RAG) based e-commerce chatbot using Ollama. The chatbot should answer FAQs, recommend products, answer product-specific questions, summarize customer reviews, and intelligently route user queries using an LLM-based intent detector.

Software Requirements:

Software	Version
Python	3.13.5
Ollama	0.40.13
VS Code	1.144.0
Pandas	3.1.0
Sentence Transformers	6.2.0
Flask	3.1.3
LLM Model	Llama 3.2 (3B) via Ollama

Dataset Description:

Dataset	Purpose
Products	products.csv – 20 products (ID, name, brand, category, price, description). Used for TF-IDF product recommendation and product-specific question answering (RAG).
FAQs	faqs.csv – 16 question/answer pairs on orders, delivery, returns, refunds, payment and warranty. Used for semantic FAQ search with embeddings and cosine similarity.
Reviews	reviews.csv – 37 customer reviews with ratings. Used for sentiment analysis, review summarization and buying suggestions.

System Architecture:

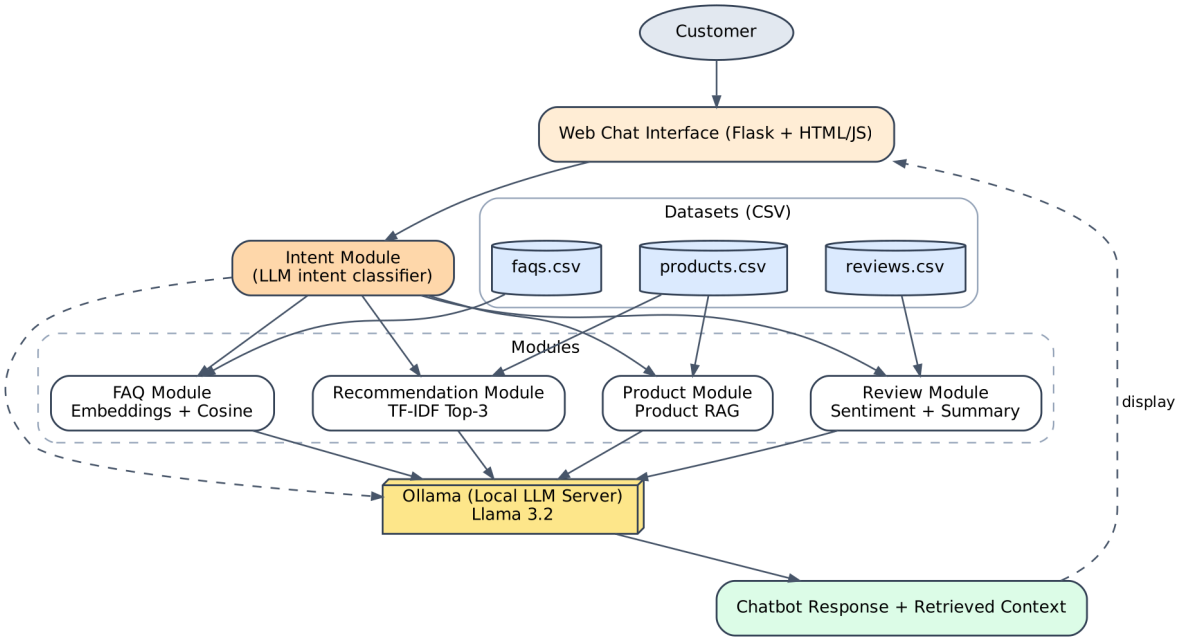


Fig. 1.1 System architecture

Module Design:

Module	Responsibility
FAQ Module	Converts FAQs into embeddings (Sentence Transformers) and finds the closest FAQ to the user query using cosine similarity. The LLM rewrites the matched answer politely.
Recommendation Module	Detects category and budget from the query, finds the top-3 similar products using TF-IDF and cosine similarity, and asks Ollama to recommend the best one.
Product Module	Retrieves the product the user is asking about and answers the question using only that product's structured information (RAG).
Review Module	Labels each review Positive/Negative/Neutral using a sentiment lexicon and generates a summary, pros, cons and a buying suggestion with the LLM.
Intent Module	Uses the LLM to classify the query as FAQ, RECOMMENDATION, PRODUCT_QA or REVIEW, with a keyword fallback, and routes the query to the matching module.
Chatbot	Integrates all modules and provides continuous chat interaction through the Flask web interface, showing the detected intent and retrieved context.

Implementation:

- GitHub Link - https://github.com/MONICK10/GPT4-Exp1-ECommerce-Chatbot
- LinkedIn Post Link - <add LinkedIn post link in config.json>
- Run: `python app.py` → `http://localhost:5001`

Output Screenshots:

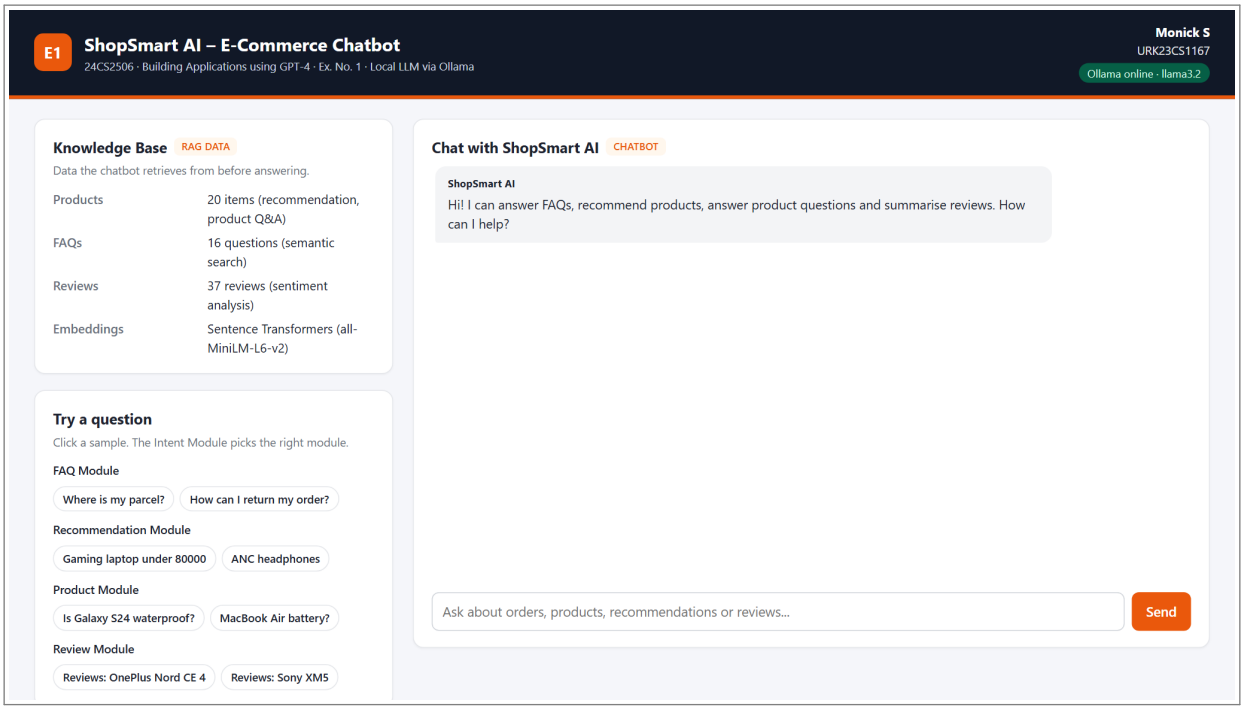


Fig. 1.2 Home Screen

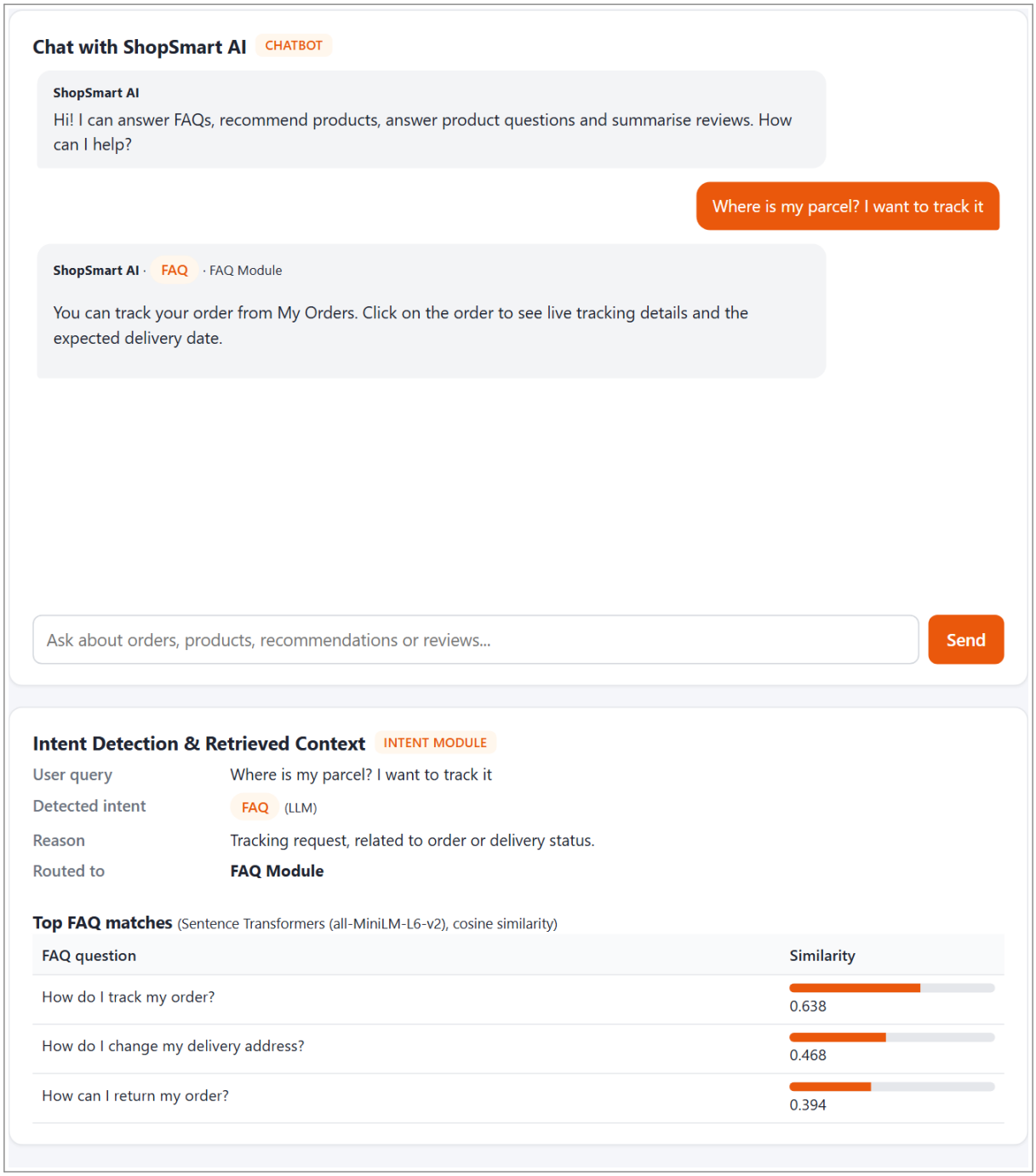

Chat with ShopSmart AI
CHATBOT
ShopSmart AI
Hi! I can answer FAQs, recommend products, answer product questions and summarise reviews. How can I help?
Where is my parcel? I want to track it
ShopSmart AI · FAQ · FAQ Module
You can track your order from My Orders. Click on the order to see live tracking details and the expected delivery date.
Ask about orders, products, recommendations or reviews...
Send
Intent Detection & Retrieved Context
INTENT MODULE
User query
Where is my parcel? I want to track it
Detected intent
FAQ (LLM)
Reason
Tracking request, related to order or delivery status.
Routed to
FAQ Module
Top FAQ matches (Sentence Transformers (all-MiniLM-L6-v2), cosine similarity)
FAQ question
Similarity
How do I track my order?
0.638
How do I change my delivery address?
0.468
How can I return my order?
0.394

Fig. 1.3 FAQ

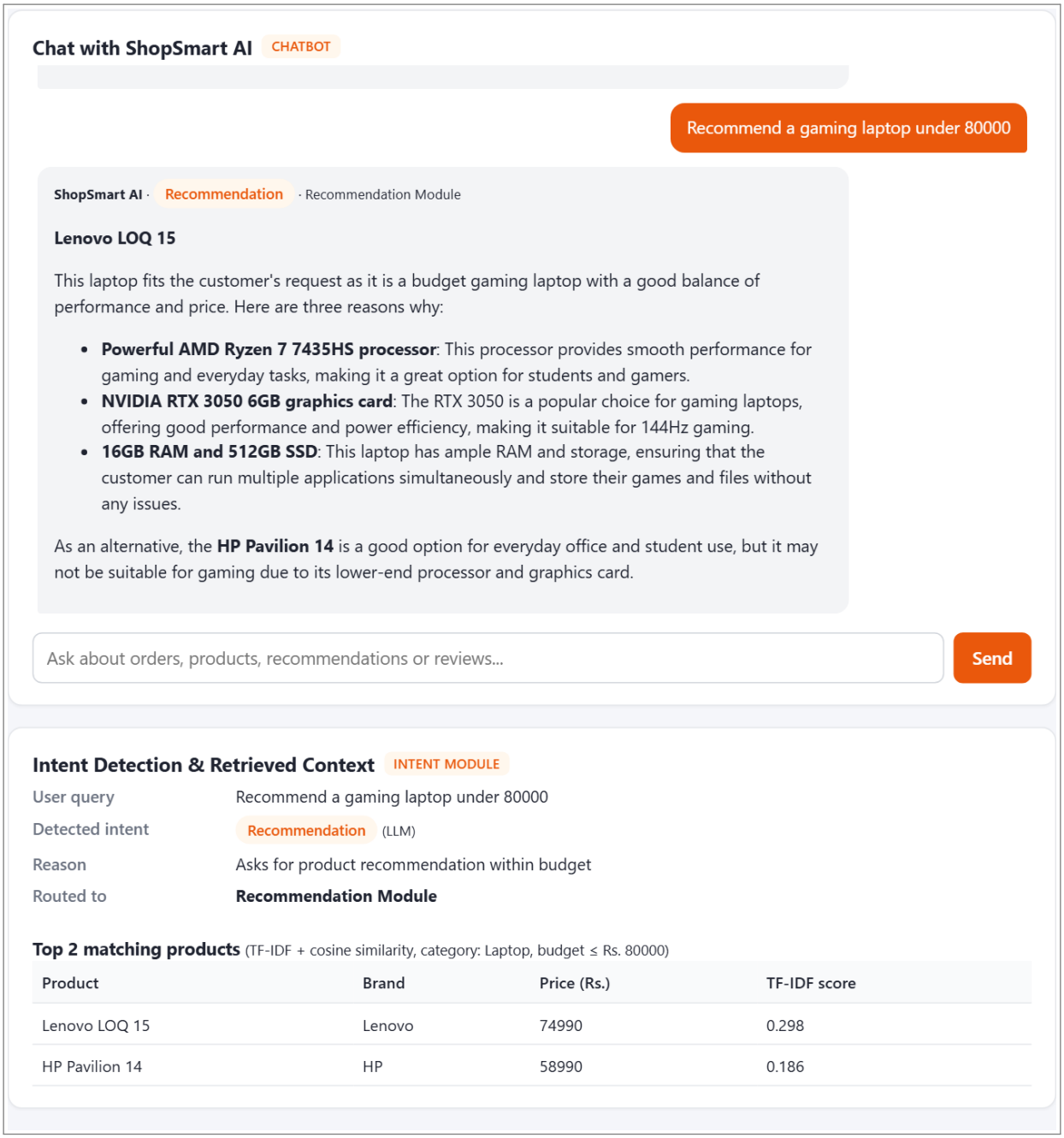


Fig. 1.4 Recommendation

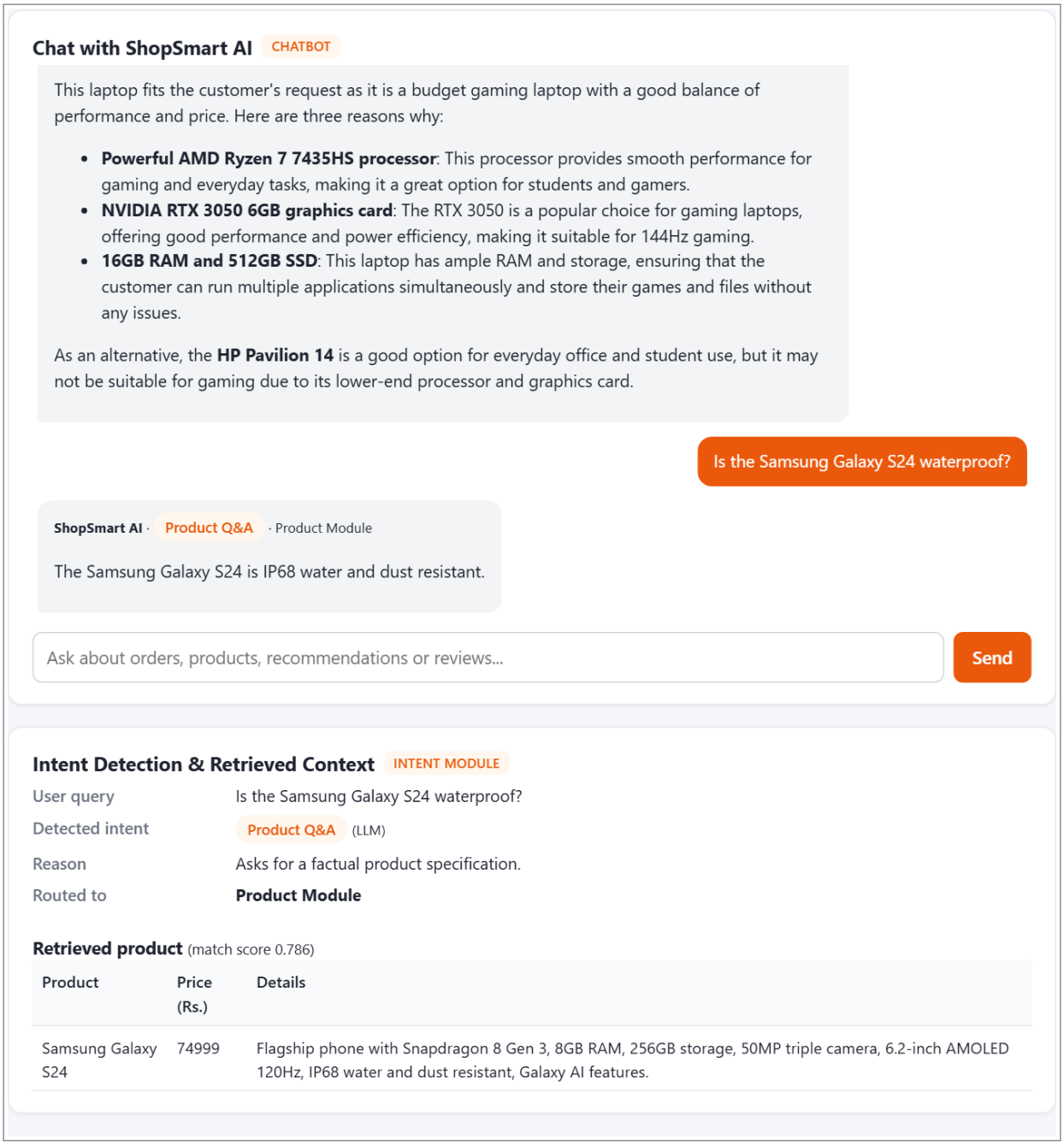


Fig. 1.5 Product QA

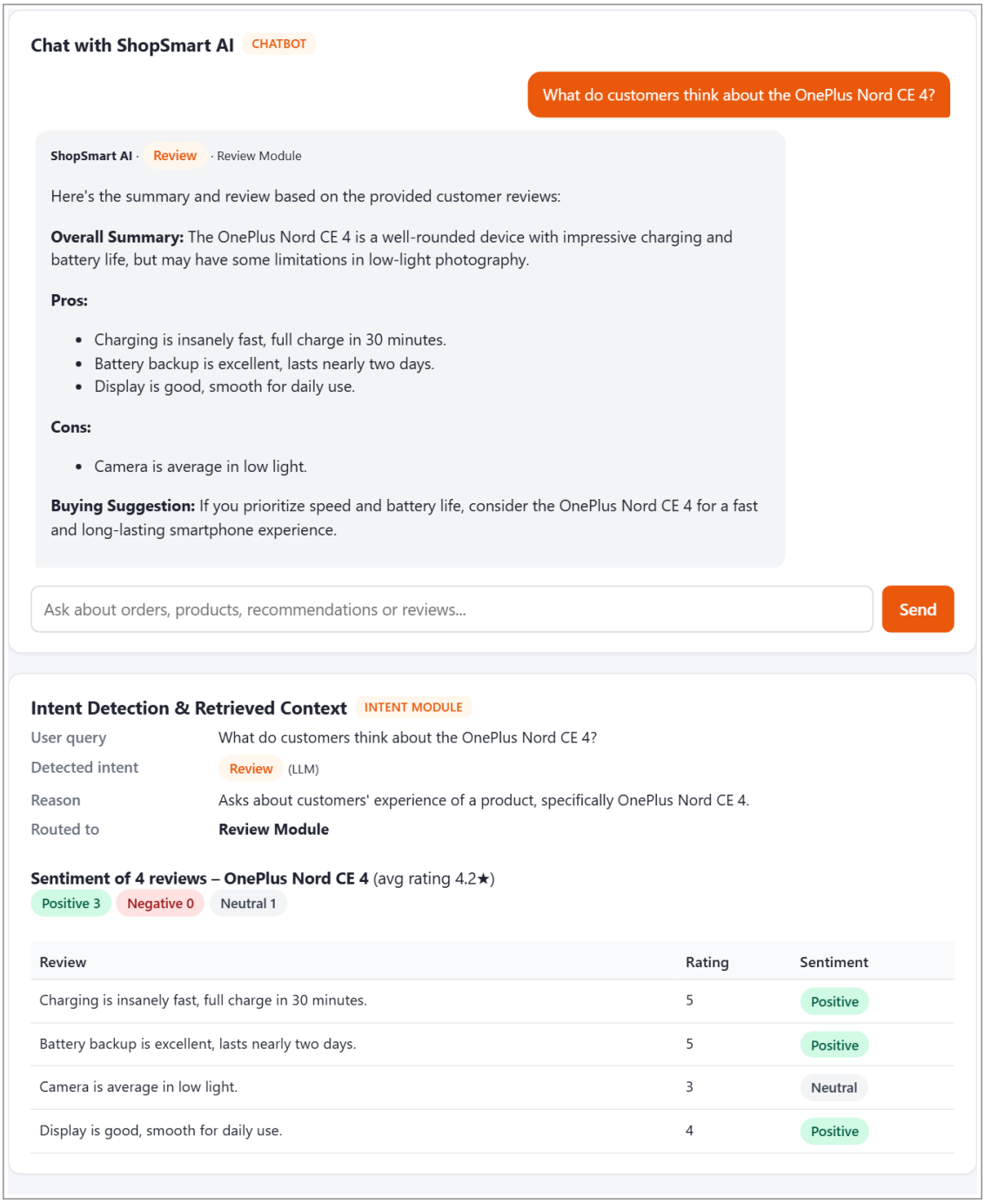


Fig. 1.6 Review Summary

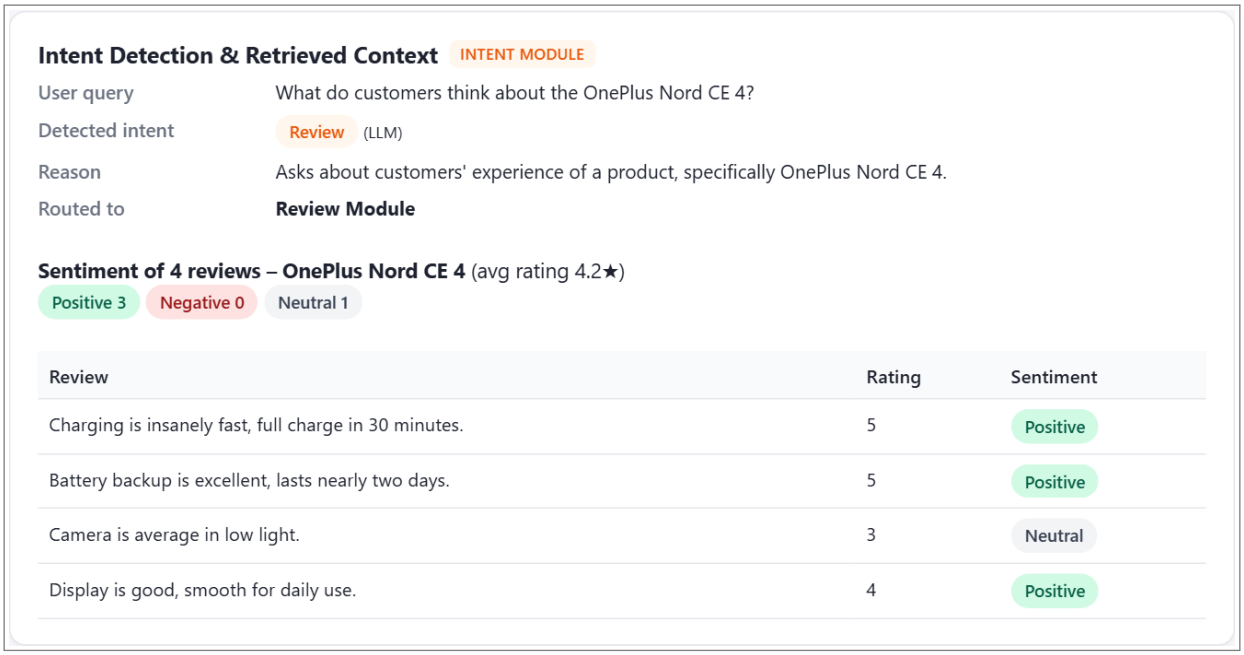


Fig. 1.7 Intent Detection

Result:

Thus successfully built a local chatbot system that can answer product-related questions from a dataset, recommend products based on similarity, analyze sentiment from user/product reviews, and simulate FAQ responses using Ollama.